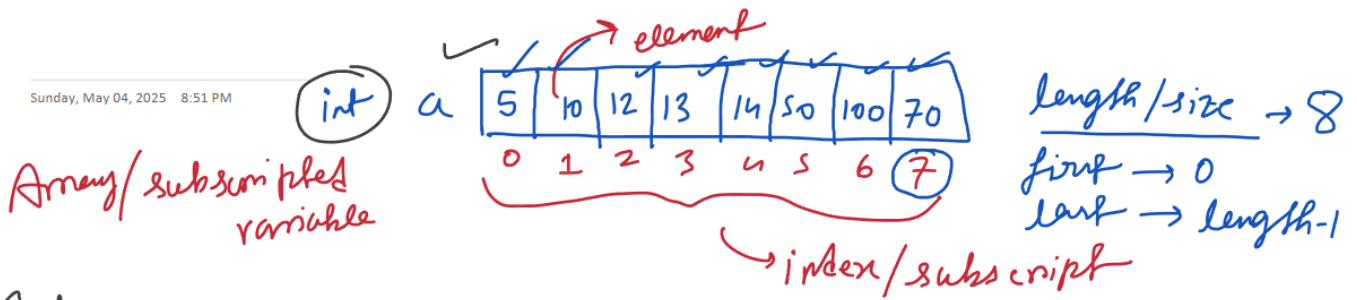




Syntax: data type array-name [] = new data type [length].

int a[] = new int [8];

a[0] = 3;

a[1] = 5;

a[2] = 7;

a[3] = 10;

a[4] = 50;

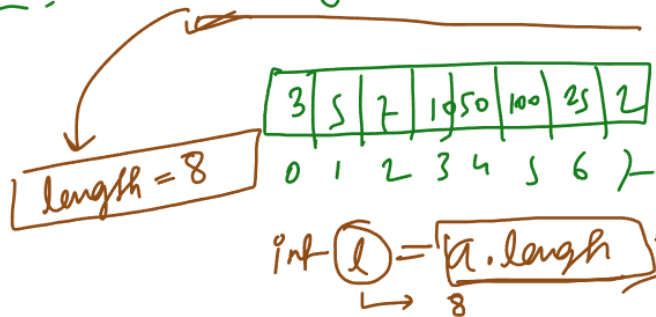
a[5] = 100;

a[6] = 25;

a[7] = 2;

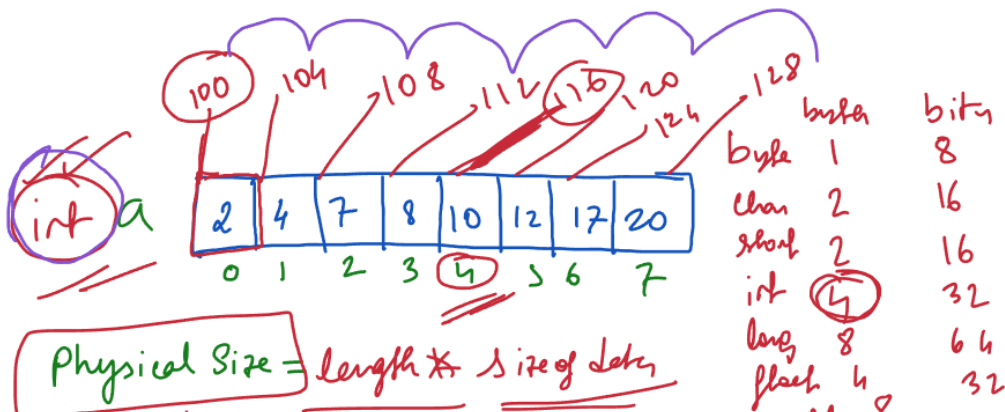
3	5	7	10	50	100	25	2
0	1	2	3	4	5	6	7

Direct Initialization: int a[] = {3, 5, 7, 10, 50, 100, 25, 2};



arr.length

b.length



Physical address of a[i] = base + (i * size of data)

$$a[4] = 100 + (4 * 4)$$

$$= 100 + 16$$

$$= 116$$

double a[] = {7.3, 5.4, 2.6};

Physical size = $3 * 8 = 24$ bytes

Traversal :



for (int i = 0; i < a.length; i++) {

 a[i] = sc.nextInt();

} Input

for (int i = 0; i < a.length; i++) {
 System.out.println(a[i]);

} Printing array

Sum of all elements :

```
int s = 0;
for (int i = 0; i < a.length; i++) {
    s += a[i];
}
```

Book : Exercise: Question 1:

```
import java.util.*;
class EvenOdd {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int a[] = new int[20];
        for (int i = 0; i < a.length; i++) {
            a[i] = sc.nextInt();
        }
        int se = 0, so = 0;
        for (int i = 0; i < a.length; i++) {
            if (a[i] % 2 == 0) {
                se += a[i];
            } else {
                so += a[i];
            }
        }
        System.out.println("Sum of Even numbers " + se);
        System.out.println("Sum of Odd numbers " + so);
    }
}
```